Question:
The y-intercepts, P and Q, of two perpendicular lines intersecting at the point $A(6,8)$ have a sum of zero. What is the area of $\triangle APQ$?
$\textbf{(A)}\ 45\qquad\textbf{(B)}\ 48\qquad\textbf{(C)}\ 54\qquad\textbf{(D)}\ 60\qquad\textbf{(E)}\ 72$

Reasoning + Answer:

<think>
1. **Identify the Coordinates:**
Given point \(A(6,8)\).

2. **Understanding the Perpendicular Lines:**
Let the slopes of the lines be \(m_1\) and \(m_2\). Since the lines are perpendicular, the product of their slopes is \(-1\):
\[
m_1 \times m_2 = -1
\]

3. **Equation of the Lines:**
The general form of the equations of the lines using point-slope form is:
\[
y - y_1 = m(x - x_1)
\]
For the first line through \(P\) (y-intercept), it can be written as:
\[
y = m_1(x - x_P) + y_P
\]
For the second line through \(Q\) (y-intercept), it can be written as:
\[
y = m_2(x - x_Q) + y_Q
\]

4. **Intercept Conditions:**
The lines pass through \(A(6,8)\). Hence:
\[
8 = m_1(6 - x_P) + y_P
\]
\[
8 = m_2(6 - x_Q) + y_Q
\]

5. **Using Perpendicularity Condition:**
Since the lines are perpendicular, using the condition \(m_1 \times m_2 = -1\), and their y-intercepts sum to zero:
\[
y_P + y_Q = 0
\]

6. **Symmetry and Equations:**
Given the symmetry and sum condition, let \(x_P = -x_Q = k\). The equations simplify since both intercepts are symmetric about the origin:
\[
y = m_1(x - k) + y_P
\]
\[
y = -m_1(x + k) + y_P
\]

7. **Slope Calculation Using Perpendicularity:**
The slopes \(m_1\) and \(m_2\) use the points. For simplicity and common point symmetry:
\[
m_1 = -\frac{1}{m_2}
\]
Hence the equations are intersecting forming \(\triangle APQ\).

8. **Finding the Area:**
Let's examine the triangle area structure. Since the y-intercepts \(P\) and \(Q\) are symmetric, and use known coordinates symmetry, which:
\[
\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}
\]
By symmetry and typical geometric area derivations:
The base can be taken from horizontal through \(A\), and height vertical through \(P\) and \(Q\).

Therefore:
Given conditions yield results easily yielding:
\[
\text{Combined area figures under corrected intersection point $A(6,8)$ yield}
\boxed{60}
</think>
<answer>
60
</answer>